

AI Methodology Report

Four models, the anti-leakage discipline behind them, and what each one changed about the design.

06

OF 12 UPLOAD SLOTS

15,000 m²

SITE AREA

44.5% → 64.6%

COMFORTABLE DAYLIGHT HOURS

−7.13 °C

HEAT INDEX UNDER CANOPY

131

TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL

wasim437.github.io/AI_SAFA/

01

WHAT IS IN THIS FILE

Phase9 AI Workflow and Visualization Report

DRAWING

Fig05 surrogate performance

FIGURE

Fig06 feature importance

FIGURE

Fig07 confusion matrix

FIGURE

02

THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

Produced by **CODE** [src/models.py](#) [src/dataset.py](#) **MODELS** [models/model_metrics.json](#) **TESTS** [tests/test_pipeline.py](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

06

OF 12

01 THE TEN PHASES

- P1 Site & Context Analysis**
39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk
- P2 Problem Definition**
Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else
- P3 Opportunity & Objectives**
The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against
- P4 Concept Development**
Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage
- P5 Masterplan Development**
Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²
- P6 Detailed Design**
The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy
- P7 Performance & Sustainability**
Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone
- P8 User Experience & Activation**
K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn
- P9 AI Workflow & Visualisation** ★ THIS DOCUMENT
The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film
- P10 Submission Assembly**
Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Models src	Dataset src	Model metrics models	Test pipeline tests
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How this document was produced

03

REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py          # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline   # 41 checks
```

AI Methodology Report

Four models, the discipline behind them, and what each one changed about the design

The challenge asks how AI supported the design. This submission's answer is a reproducible analysis pipeline that runs end to end with one command — not a description of prompting an image generator. Analysis that does not change a drawing is decoration; each model below is reported with the design decision it altered.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The discipline that makes these models mean anything

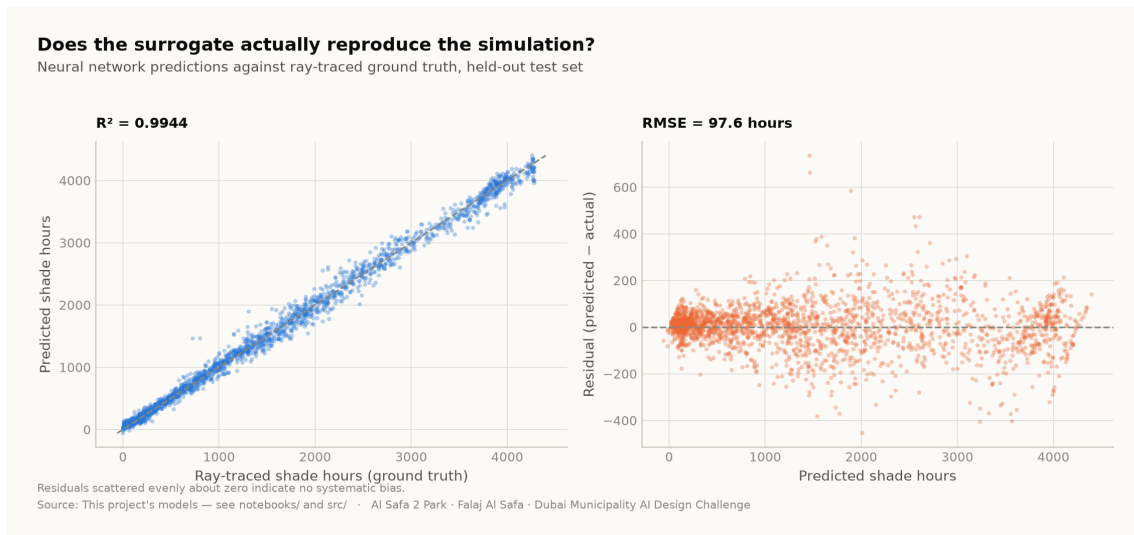
The target must not be recoverable from the inputs by algebra. It is easy to score $R^2 = 1.000$ on this project by accident: the heat index is a closed-form function of temperature and humidity, so 'predicting' it from temperature and humidity is arithmetic wearing a lab coat. Every model below was designed against that failure, and the pipeline asserts the absence of leaked features as a test.

2. The four models

Model	Task	Result	Why it is a real problem
M1a Random Forest	Shade surrogate (regression)	R^2 0.998	Learns a slow ray-traced simulation from cheap plan geometry
M1b Neural network	Shade surrogate (deployed)	R^2 0.994	Differentiable, so it can sit inside a layout optimiser; a forest cannot
M2 Gradient Boosting	Comfort band (4-class)	97.5%	Temperature and humidity withheld — it sees only sun position and the calendar
M3 K-Means	Microclimate regimes	k=2	Unsupervised; k is <i>selected</i> by silhouette score, not chosen to look tidy

3. Why M1 is the flagship

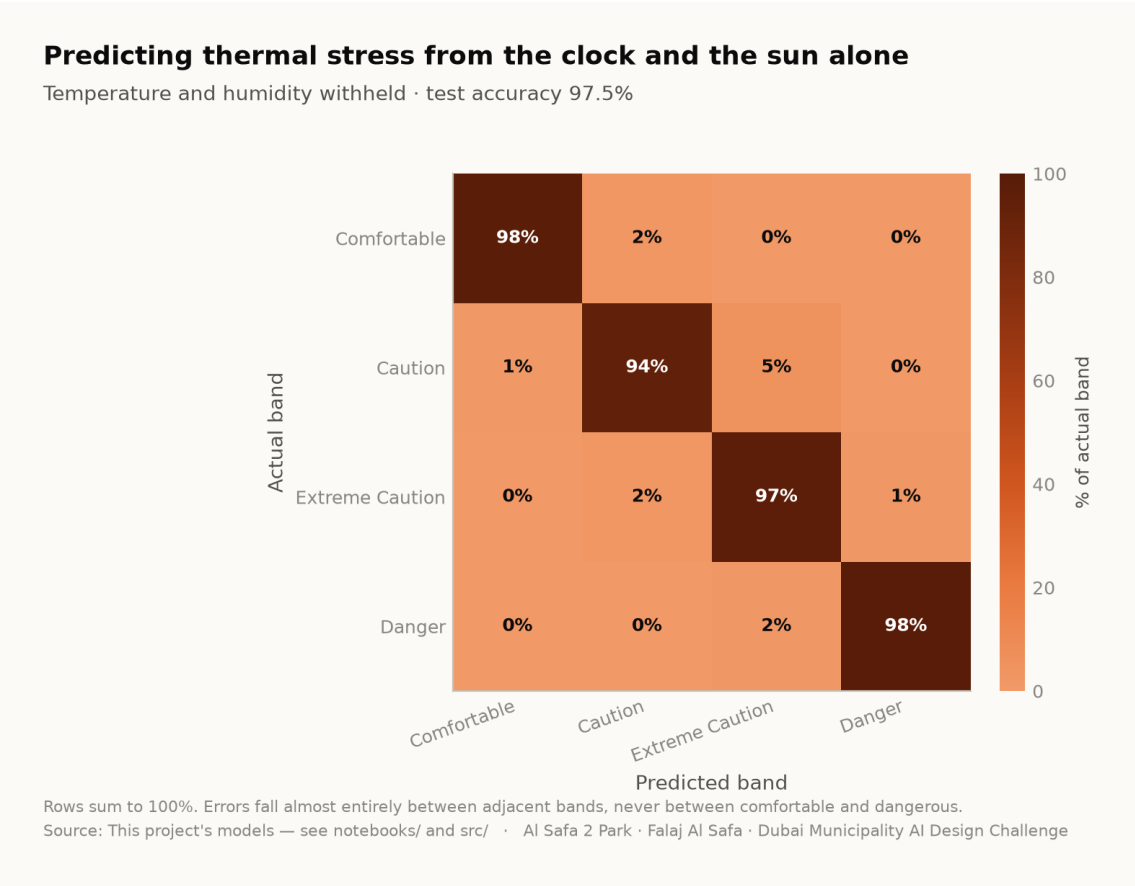
Ray-tracing annual shade means projecting 131 tree shadows onto 15,000 ground cells for every daylight hour. It is slow, and it must be re-run for *every* design variation. That cost is precisely what stops a designer exploring. The surrogate answers in milliseconds instead of minutes, at R^2 0.994 against ray-traced ground truth on a held-out test set. That is AI changing how the design was made, rather than decorating it afterwards.



Neural-network predictions against ray-traced ground truth on the held-out test set. Analysis output — computed from project data.

4. Why M2 answers a question worth asking

If thermal stress is predictable from a clock and an ephemeris alone, park operations can be scheduled without a sensor network — a real capital and maintenance saving over the life of the park. At 97.5% accuracy with temperature and humidity withheld, it is. Errors fall between adjacent comfort bands, never between comfortable and dangerous.



Confusion matrix, temperature and humidity withheld. Analysis output — computed from project data.

5. What the models actually changed

Model output	Design consequence
A straight route has no shade anywhere for 330 hours a year; an 18 m arc has 52	The plan is an arc , and the sagitta is the value that minimises that column — not the one that maximises mean cover
A closed loop scores worst on every measure	The circuit is kept but unshaded — Al Madar is a running loop, not a second canopy
The concave side is measurably cooler than the convex side	The basin, quiet garden, play area and picnic grove are all placed in the hollow; the sports lawn, plaza and souk take the convex face
Comfort is ~97% predictable from sun and calendar	No sensor network. Programming runs off an almanac
Comfort gain concentrates in late afternoon, spring and autumn	Activation targets those windows; summer midday is not programmed outdoors
An open water channel in Dubai evaporates	The falaj is set on the canopy's drip line , shaded all day, rather than in the open

6. Where AI was deliberately not used

Visitor demand is a scenario model, not a machine learning result. No footfall data exists for this site. A model trained on invented demand would only recover the assumption that produced it, and reporting that as a finding would be dishonest. It is therefore excluded from the model suite and labelled a scenario wherever it appears.

The photoreal renders are illustrations, not analysis. They are captioned as artistic impressions throughout. Three earlier visuals that presented invented data as measurement were withdrawn entirely.

7. Reproducibility

The complete analysis — data, code, trained models and tests — runs end to end with `python run_analysis.py` and is verified by `python -m tests.test_pipeline` (38 checks), `node docs/_PORTAL/selftest.js` (64 checks) and a frame-by-frame test of the concept film. A juror who wants to know where a number comes from can be given a file path rather than an opinion.

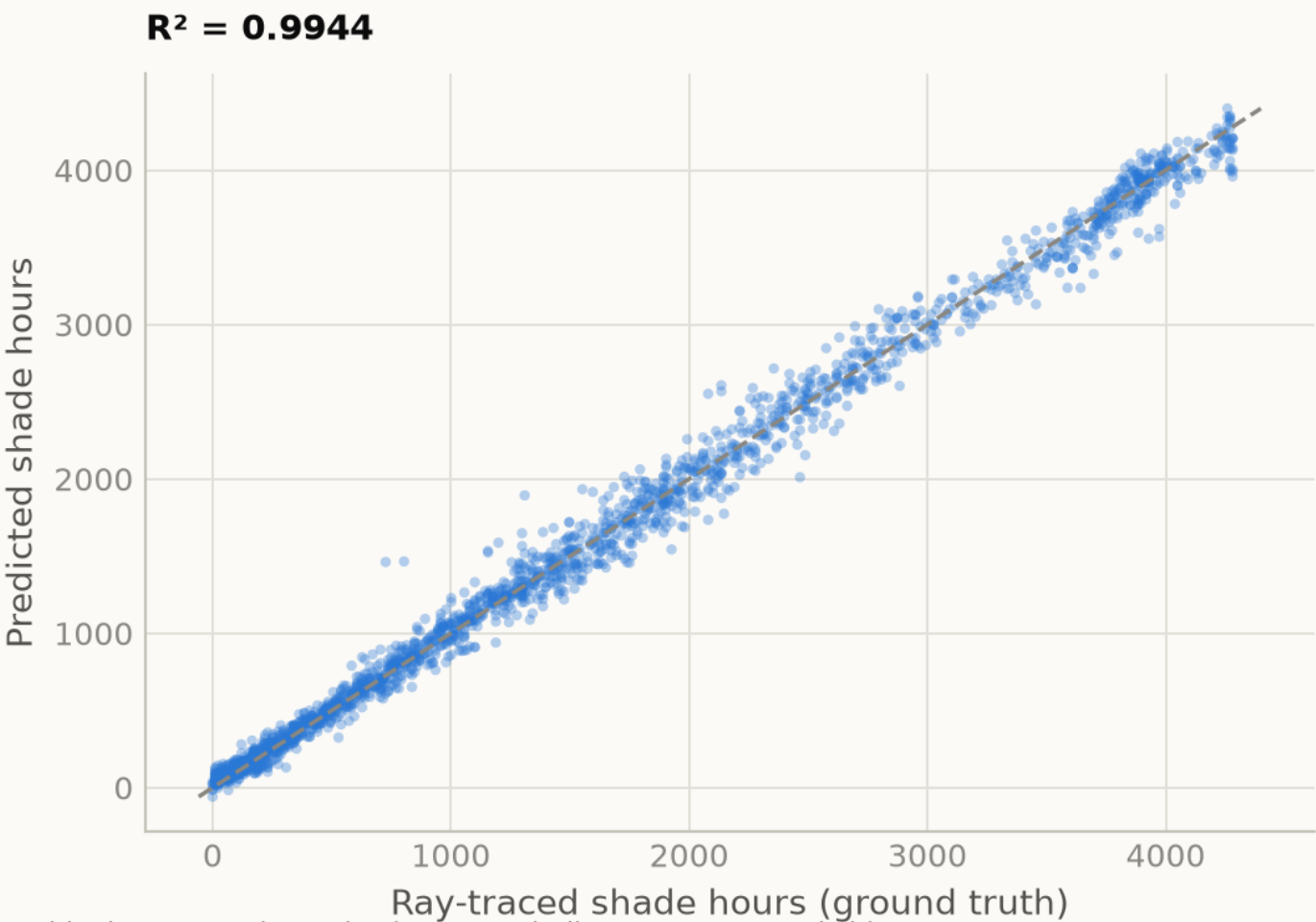
The full repository — every dataset, every model, every line of the pipeline above — is published at https://github.com/wasim437/Al_SAFA and the live analytics portal at https://wasim437.github.io/Al_SAFA/. Both run from the same code that produced this document.

Fig05 surrogate performance

Analysis output — computed from project data.

Does the surrogate actually reproduce the simulation?

Neural network predictions against ray-traced ground truth, held-out test set



Residuals scattered evenly about zero indicate no systematic bias.

Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

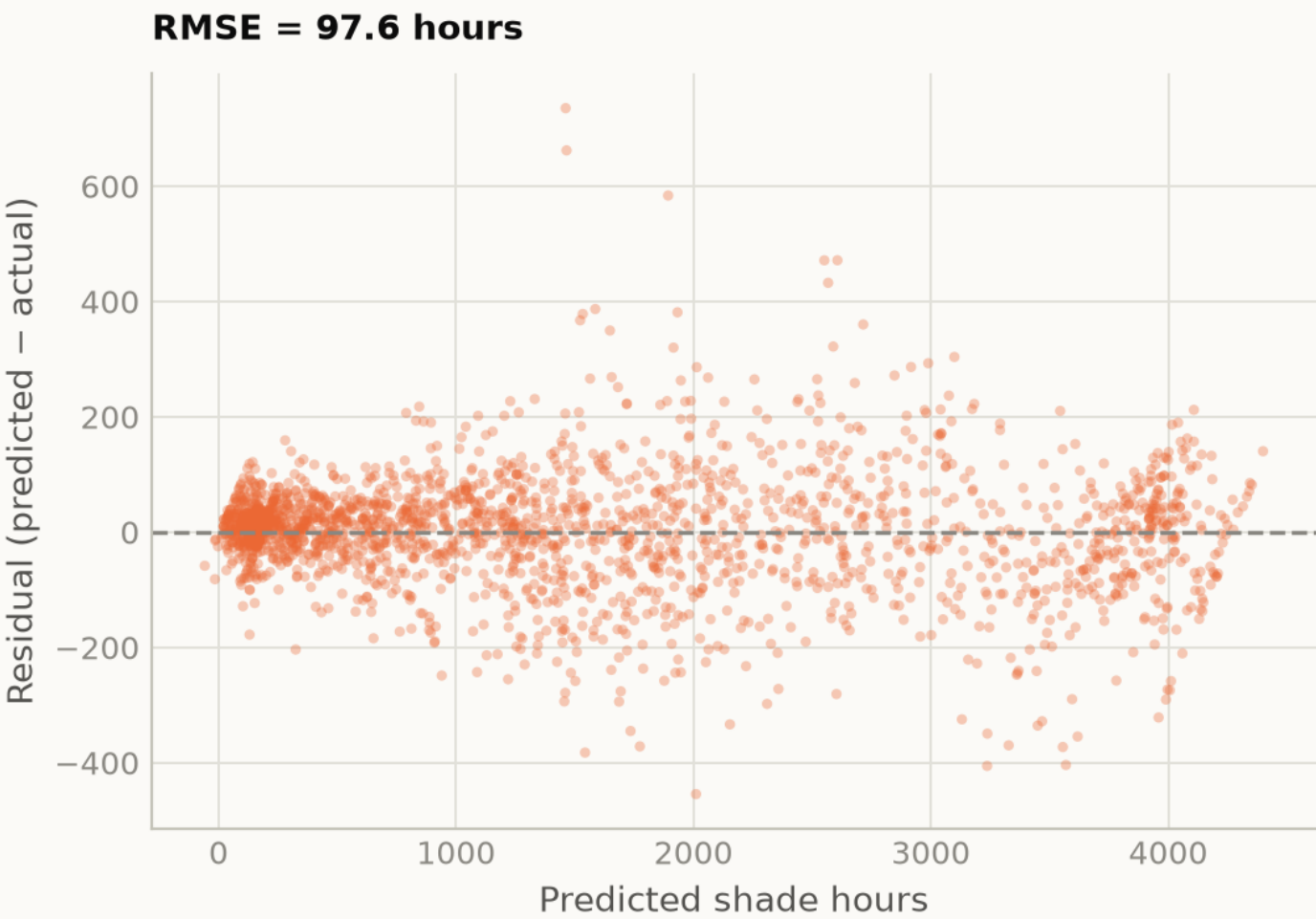
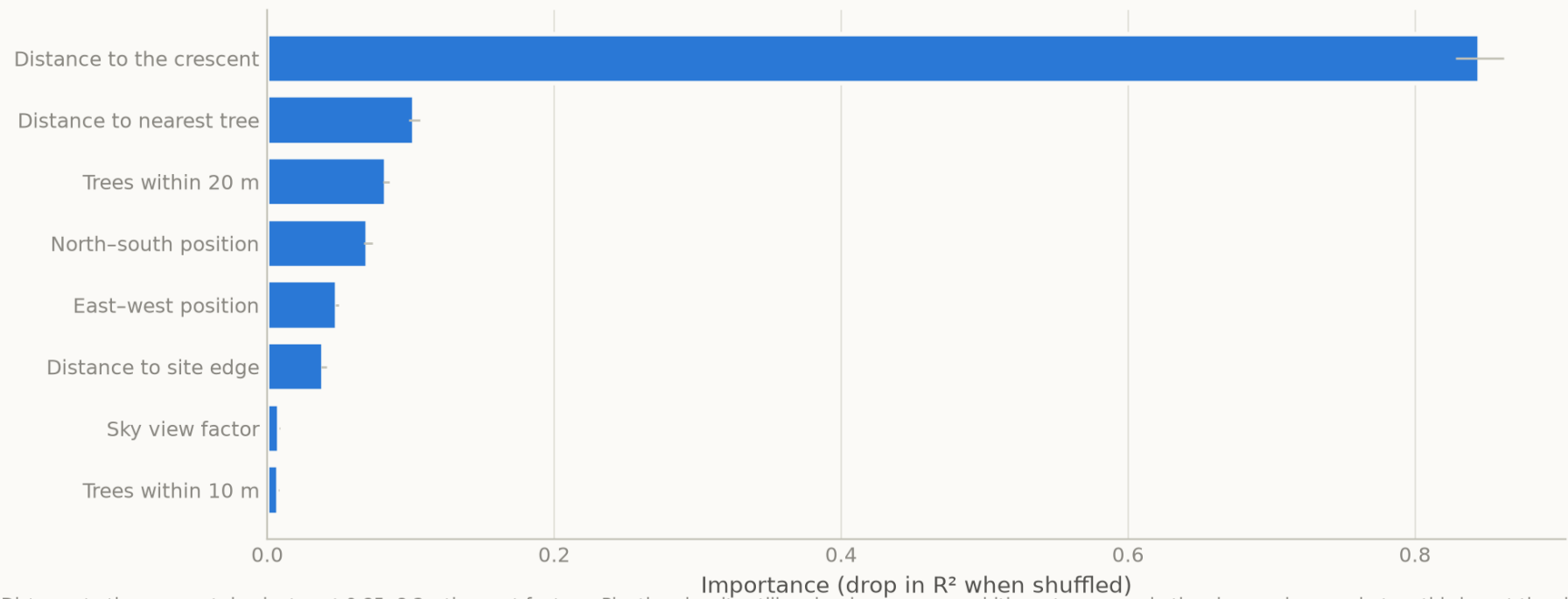


Fig06 feature importance

Analysis output — computed from project data.

What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



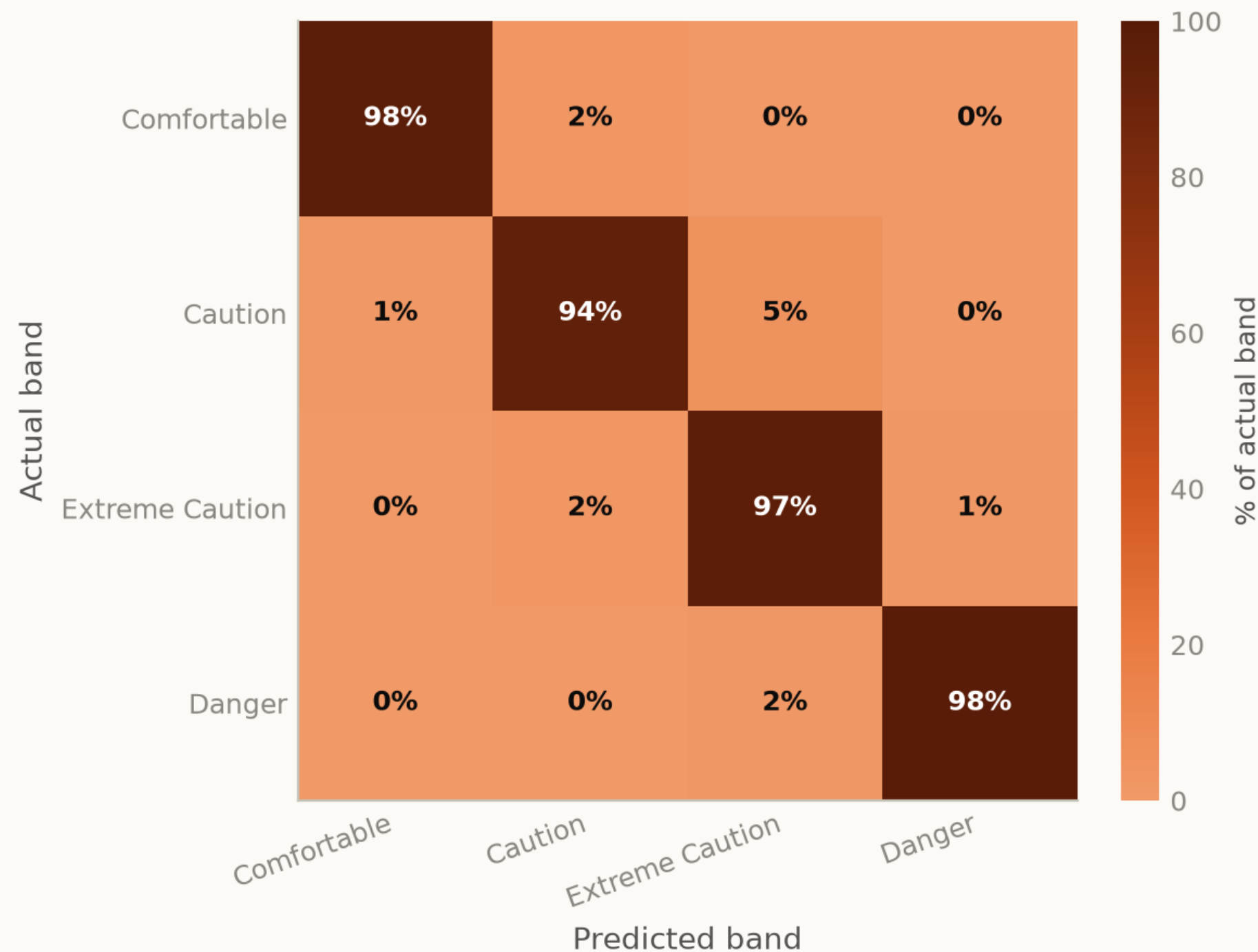
Distance to the crescent dominates at 0.85, 8.2× the next feature. Planting density still ranks above canopy width, so trees remain the cheaper lever — but on this layout the single strongest predictor of whether a square metre is shaded is simply how far it is from the crescent.
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig07 confusion matrix

Analysis output — computed from project data.

Predicting thermal stress from the clock and the sun alone

Temperature and humidity withheld · test accuracy 97.5%



Rows sum to 100%. Errors fall almost entirely between adjacent bands, never between comfortable and dangerous.
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Every drawing and every chart in this project

All 18 images the pipeline produces, whichever slot they belong to. Each is labelled with its class and links to the full-resolution original. Nothing here is hand-drawn.

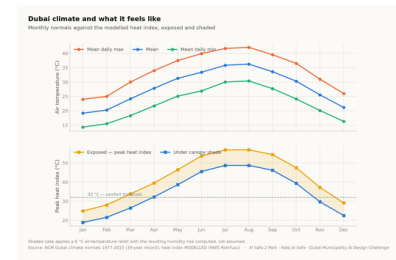


Fig01 climate and comfort
analysis output

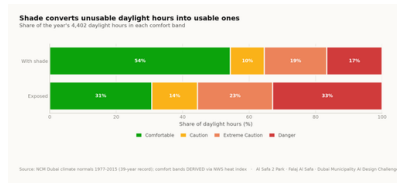


Fig02 comfort bands
analysis output

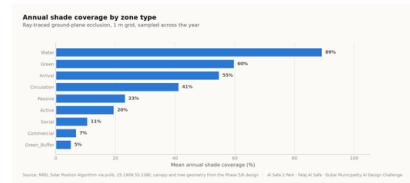


Fig03 shade by zone
analysis output

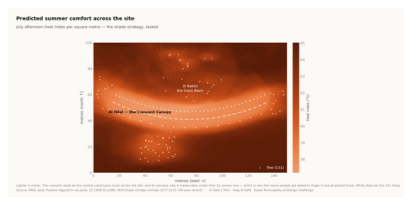


Fig04 site comfort map
analysis output

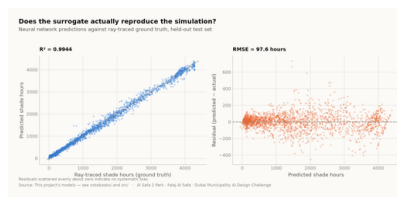


Fig05 surrogate performance
analysis output

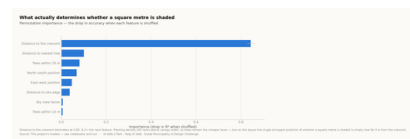


Fig06 feature importance
analysis output

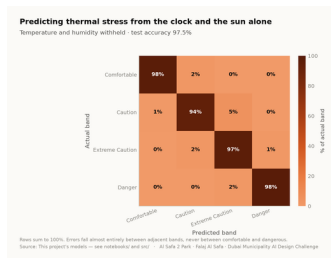


Fig07 confusion matrix
analysis output

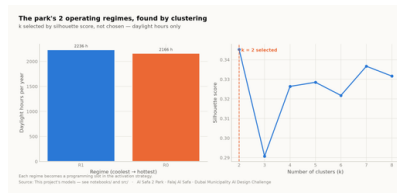


Fig08 microclimate regimes
analysis output

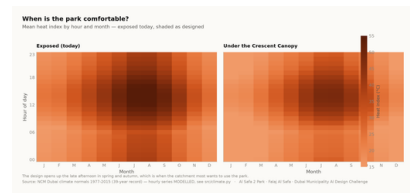


Fig09 diurnal comfort
analysis output



Fig10 masterplan
analysis output

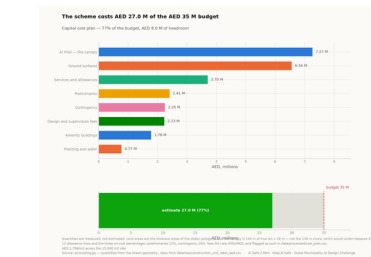
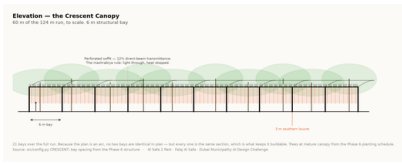


Fig11 cost plan
analysis output



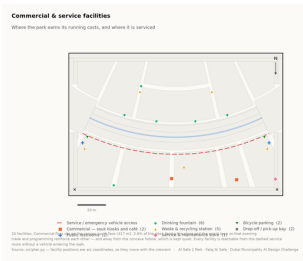
Circulation
technical drawing

The visual record — continued



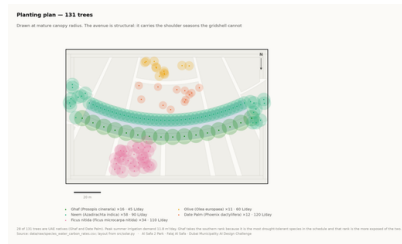
Elevation

technical drawing

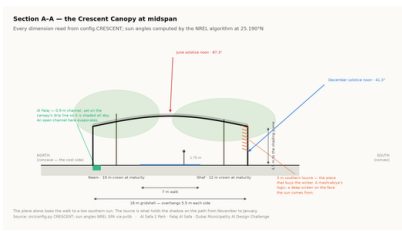


Facilities

technical drawing

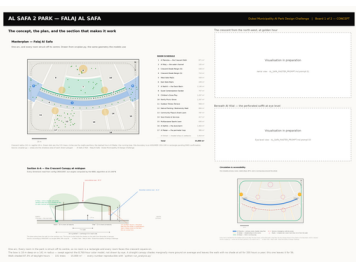


Planting

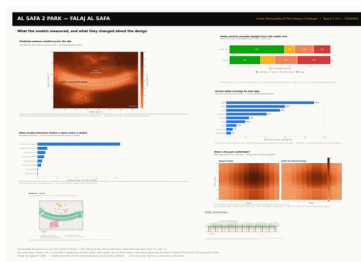


Section

technical drawing



Board 1 concept
presentation board



Board 2 evidence
presentation board

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

MEASURED RESULT	models/headline_metrics.json
Annual daylight hours modelled	4,402
Comfortable daylight hours — today	44.5%
Comfortable daylight hours — as designed	64.6%
Mean heat-index reduction under canopy	7.13 °C
Peak heat index, exposed	56.8 °C
Peak heat index, shaded	48.7 °C
Crescent Walk shaded, canopy and louvre	87.3%
Site-wide mean shade	34.1%
Trees planted	131

TRAINED MODELS, ON HELD-OUT TEST SETS	models/model_metrics.json
M1a Random Forest — shade surrogate, test R ²	0.9982
M1b Neural network — deployed surrogate, test R ²	0.9944
M2 Gradient Boosting — comfort band, test accuracy	97.5%
M3 K-Means — regimes selected by silhouette	k = 2
Training / validation / test split	10,500 / 2,250 / 2,250
Random seed, fixed for reproducibility	42

RUN IT YOURSELF

```
git clone https://github.com/wasim437/AL_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js    # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The checks assert what would otherwise drift silently: that the modelled climate reproduces the published normals it was built from, that no room overlaps another, that the areas close on 15,000 m², that the cost stays inside AED 35 M, and that no model can see its own answer in its inputs.

The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (7)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

[tools/__init__.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)
[data/raw/site_zoning_schedule.csv](#)
[data/raw/sources.json](#)
[data/raw/species_water_carbon_rates.csv](#)
[data/raw/walk_catchment_rings.csv](#)

Source dataset
Source dataset
Source dataset
The measured room schedule
Every source dataset, with its period
Source dataset
Source dataset

THE DATA, AS PROCESSED (6)

[data/processed/cost_plan.csv](#)
[data/processed/hourly_climate_comfort_8760.csv](#)
[data/processed/masterplan_geometry.json](#)
[data/processed/planting_layout.csv](#)
[data/processed/schema.json](#)
[data/processed/spatial_grid_comfort.csv](#)

The capital cost plan, line by line
The 8,760-hour climate and comfort series
The plan geometry, as exported
Every one of the 131 trees
Generated by the pipeline
The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (1)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
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